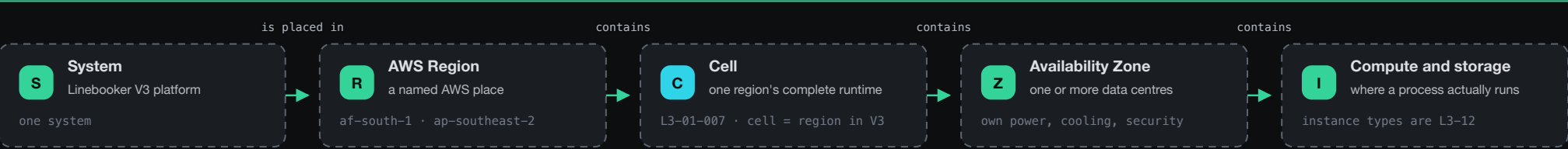


Where Every Running Thing Sits

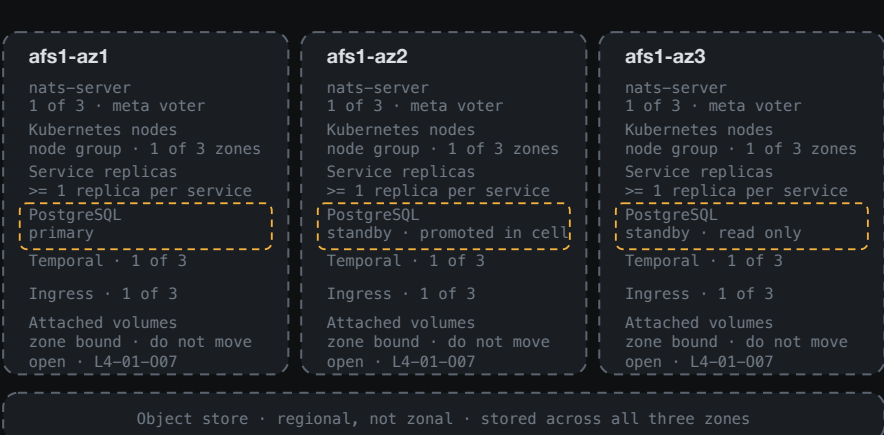
Where does every running thing sit, from the system down to the availability zone, and what runs beside that placement? Defers: instance types, node counts and sizing (LB-V3-L3-12); cluster shape (sheet 2); cost (sheet 3). **This is a directional document:** a first-pass position as at 18 September 2026. Nothing on this sheet is built.

PHYSICAL CHAIN · EVERY ELEMENT ON THIS DOCUMENT HAS ONE HOME IN IT



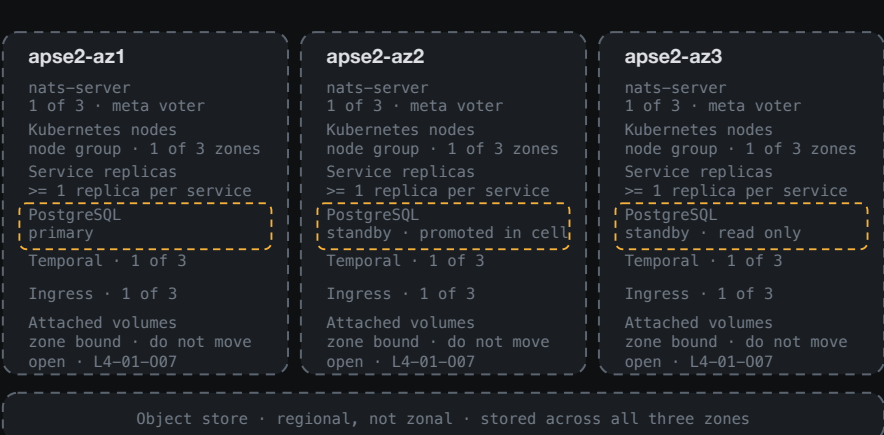
AWS REGION AF-SOUTH-1 · AFRICA (CAPE TOWN) · 3 ZONES · OPT-IN

CELL ZA · ONE REGION'S COMPLETE RUNTIME

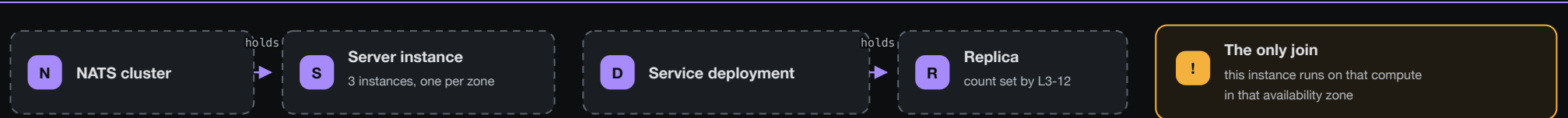


AWS REGION AP-SOUTHEAST-2 · ASIA PACIFIC (SYDNEY) · 3 ZONES · ON BY DEFAULT

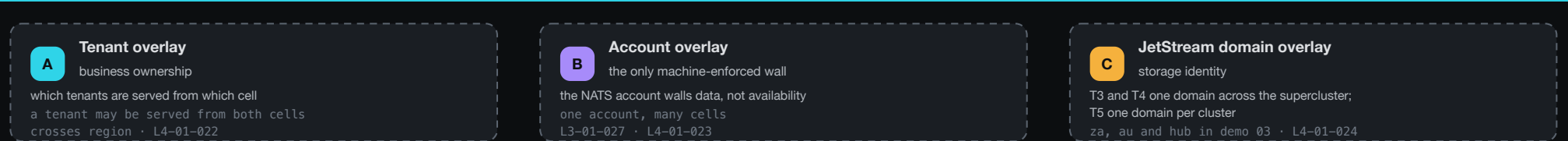
CELL AU · ONE REGION'S COMPLETE RUNTIME



RUNTIME CHAIN · DRAWN BESIDE THE PLACEMENT, NOT INSIDE IT



OVERLAYS · FIVE AXES CROSS, THEY DO NOT NEST (ADR-056, L3-01-010)



Read it downward. The physical chain names one home for everything; the two cells fill it in; the runtime chain is drawn beside it because a cluster is not a place; the overlays cross both. **Built today:** nothing — the lab runs one server in one region. **Not shown:** instance types and sizing, which are LB-V3-L3-12 ; cluster shape, which is sheet 2; cost, which is sheet 3.

SCOPE	One question. Where does every running thing sit? Cluster shape is sheet 2, cost is sheet 3.
DIRECTION	Directional. A first pass as at 2026-09-18 so work can start. Not a decision and not a forecast.
CELL	In V3 the cell boundary and the AWS Region boundary are the same boundary (L4-01-002).
ZONE	An availability zone is one or more data centres with their own power, cooling and physical security. It is not one building (L4-01-005).
ZONE ID	Zones are named by zone ID, afs1-az1 , not af-south-1a . The account-local name is shuffled per account; the ID is not (L4-01-006).
OPT-IN	af-south-1 is opt-in. An account must enable it before anything can be placed there. ap-southeast-2 is on by default (L4-01-003 , L4-01-004).
THREE	Three NATS servers across three zones is the smallest number that keeps a majority when one zone is lost (L4-01-008).
ONLY NATS	Three NATS servers prove NATS availability only. Every other stateful part states its own replica count and zone spread (L4-01-010).
BESIDE	Placement and runtime are two chains, not one tree. The only join is: this instance runs on that compute in that zone (L4-01-007).
OPEN	Zone spread and failover for Postgres, the object store, Temporal, ingress, nodes and volumes is open. See L4-01-007 , owned by Architecture.

LEGEND

	Green boundary — physical geography: zone, cell, region, country		Cyan boundary — the cell boundary and what crosses it		Purple boundary — runtime: clusters, instances, deployments, replicas		Dashed grey tile — proposed for V3, not built
	Dashed amber tile — open, a named owner must confirm		Amber panel — a rule or a rate that governs the sheet		Green line — a containment or placement relationship		Purple line — a runtime relationship
	Arrow head — the direction of the relationship	Line label — names the relationship, never bare "uses"					

WORDS ON THIS SHEET

AWS Amazon Web Services	meta group the servers that vote on JetStream changes
AZ availability zone, one or more data centres with their own power	replica one more copy of the same thing
cell one region's complete runtime	JetStream domain the storage identity of a cluster
region a named AWS place holding several zones	ADR architecture decision record
zone ID the stable zone name, afs1-az1	NATS the messaging system; a product name, not an abbreviation
opt-in a region an account must enable before use	
RAFT the vote NATS uses to pick a leader	

Cluster Shapes T3, T4 and T5

Which clusters exist, who votes, and what does each shape survive? Defers: which shape is chosen (L4-01-001, open, Architecture); cost per byte, which is sheet 3. **This is a directional document:** a first-pass position as at 18 September 2026. Nothing on this sheet is built.

CLUSTER IS THE AXIS THAT SEPARATES THE THREE SHAPES (L4-01-009)

T3 · GATEWAY SUPERCLUSTER · ONE ARBITER

z

za cluster

3 servers · af-south-1

gateway · one supercluster

a

au cluster

3 servers · ap-southeast-2

gateway · one supercluster

v

arbiter

1 server · holds no stream assets

7 meta voters · majority 4

one meta group · one JetStream domain

lose one region: 4 voters left, a majority with zero slack

evidence: demo 03 checks C1 to C12

z

za cluster

3 servers · af-south-1

gateway · one supercluster

a

au cluster

3 servers · ap-southeast-2

gateway · one supercluster

v

arbiter cluster

3 servers · holds no stream assets

9 meta voters · majority 5

one meta group · one JetStream domain

lose one region: 6 voters left, one node of slack (F11)

survives a region plus one more node, and no further (F12)

evidence: demo 03 checks F1 to F12

h

hub cluster

3 servers · af-south-1 · domain hub

decision taken 2026-09-18 · L4-01-016

leafnode Link

z

za leaf

3 servers · af-south-1 · domain za

leafnode Link

a

au leaf

3 servers · ap-southeast-2 · domain au

three separate meta groups of 3 · majority 2 each

one JetStream domain per cluster (za, au, hub)

lose the whole hub: both leaves keep a meta leader and still create streams (D10 to D14)

nothing replicates between domains by itself (D15)

evidence: demo 03 checks D1 to D15

COMPARED ON ONE BASIS (L4-01-014)

	T3	T4	T5
Meta voters	7	9	3 + 3 + 3, separate
Majority	4	5	2 in each group
Survives one zone	yes	yes	yes
Survives one region	yes, zero slack	yes, one node of slack	yes, each leaf on its own
When the majority is lost	no new stream or consumer	no new stream or consumer	one group at a time; the other two carry on
Cost to move a byte	gateway traffic crosses the border	same, plus a third region's control traffic	au-to-hub crosses a region; za-to-hub does not

LOSING A MAJORITY BLOCKS CHANGES, NOT TRAFFIC (L4-01-015)

Creating or editing a stream or consumer returns 10008 JetStream system temporarily unavailable	Only after the old leader ages out, about 40 seconds (D03-R5, F7a)	Publishing into a stream that already exists keeps working (F9)
--	---	--

One axis, three answers. Cluster is the only axis that separates these shapes; placement and overlays are the same in all three. **Built today:** nothing. **Measured:** the voter arithmetic only, as bare `nats-server` processes on one Mac. **Not shown:** the chosen shape — that is open item L4-01-001 .

SCOPE

One question. Which clusters exist, who votes, what survives? Cost is sheet 3.

DIRECTION

Directional as at 2026-09-18. No V3 decision of record exists (L3-08-010).

AXIS

Cluster is the axis that separates the shapes. Region, tenant, account, cell and domain are the same in all three (L4-01-009).

GATEWAY

A gateway joins two clusters into one supercluster with one meta group (T3, T4).

LEAFNODE

A leafnode link joins a cluster to a hub and keeps its own meta group and its own domain (T5).

ARBITER

The arbiter holds no stream assets. It votes, and that is all (L4-01-030).

HUB

If T5 is chosen the hub sits beside the `za` cell in `af-south-1` . One region failure then takes the hub and the `za` cell together; the `au` leaf keeps serving (L4-01-016 , L4-01-017).

EVIDENCE

Demo 03 ran these shapes as bare processes on one Mac. It measured NATS quorum, not AWS (L4-01-031).

OPEN

Which shape is chosen is open. It is decided by which failure boundary the business is buying (L4-01-001 , owned by Architecture; L4-01-003 , owned by the business owner).

LEGEND

Blue boundary

— a gateway supercluster

Red panel

— a warning or a limit on the evidence

Line label

— names the relationship, never bare "uses"

Purple boundary

— runtime: clusters, instances, deployments, replicas

Blue line

— a gateway link between clusters

Amber boundary

— cost, rates and rules that govern the sheet

Cyan dotted line

— a leafnode link, one thing that crosses a cell boundary

Dashed grey tile

— proposed for V3, not built

Arrow head

— the direction of the relationship

WORDS ON THIS SHEET

AWS

Amazon Web Services

cell

one region's complete runtime

RAFT

the vote NATS uses to pick a leader

meta group

the servers that vote on JetStream changes

quorum / majority

more than half the voters

gateway

a link joining two NATS clusters into one supercluster

Leafnode

a link joining a cluster to a hub, keeping its own meta group

JetStream domain

the storage identity of a cluster

arbiter

a voter holding no stream data

mirror

a copy of one stream in another place

replica

one more copy of the same thing

ADR

architecture decision record

NATS

the messaging system; a product name, not an abbreviation

Traceability: L4-01-009 and L4-01-011 to L4-01-017 and L4-01-030 (cluster axis, the three shapes, the one-basis comparison, quorum-loss behaviour, the hub decision). Evidence: demos/03-multi-cluster-and-accounts/REPORT.md checks C1-C12, F1-F12, D1-D15. Open: L4-01-001 (Architecture), L4-01-005 (Architecture).

LB-V3-L4-01 · 2/3 · CLUSTER SHAPES

Failure Boundaries and Transfer Cost

Which failure is being bought, and what does each shape cost per byte moved? Defers: the actual GB per day per cell (L4-01-004, open) — every figure here is a shape, not a forecast. **This is a directional document:** an initial estimate as at 18 September 2026. Nothing on this sheet is built.

THREE FAILURE BOUNDARIES, TREATED SEPARATELY (L4-01-018)

1 One availability zone

a zone is lost

three servers across three zones keep a majority
answered by all three shapes

CANDIDATE ARBITER HOME

M

ap-southeast-4 · Asia Pacific (Melbourne)

3 zones · opt-in

a second AWS Region in Australia, recorded as a
candidate arbiter home

it is not a cell (L4-01-019)

2 One AWS Region

a whole region is lost

needs voters outside that region

T3 zero slack · T4 one node of slack ·
T5 each leaf on its own

3

One country or jurisdiction

a country is lost or closed

needs a third jurisdiction, not a third region
open · L4-01-003 · business owner

!

Vote-bias analysis

Sydney and Melbourne are separate AWS Regions in ONE country (L4-01-020)

In T3 an arbiter in Melbourne gives Australia 3 + 1 = 4 of 7, a majority.
South Africa holds 3 and can never reach one (L4-01-021)

An arbiter in Melbourne answers an AWS Region failure. It does not answer a country failure.

TRANSFER COST IS A DESIGN INPUT, NOT A FOOTNOTE (L4-01-025)

Rate card · AWS list prices, published 2026-09-16, version 20260916132208

Inside one zone, private addresses

free

Across a zone, same region

\$0.01 / GB, charged at BOTH ends = \$0.02 / GB

Same rate in af-south-1, ap-southeast-2 and ap-southeast-4

af-south-1 → ap-southeast-2

\$0.147 / GB

ap-southeast-2 → af-south-1

\$0.098 / GB

ap-southeast-4 → af-south-1

\$0.100 / GB

ap-southeast-2 ↔ ap-southeast-4

\$0.080 / GB

cross-region is charged on the SENDING side only; inbound is free

Formula (L4-01-028)

cost per cell per day = D x ((R - 1) + f) x \$0.02

D = GB published per day · R = JetStream replica count

f = share of reads served from a different zone
with R = 3 and f = 0.67 the factor is 2.67

One cross-border GB costs about SEVEN cross-zone GB (\$0.147 against \$0.02)

The zone spread is cheap. The region crossing is the
part worth designing around (L4-01-029).

Worked table · 30-day month

Published per cell per day	Cross-zone, one cell	Mirror it all Cape Town to Sydney	Mirror it all Sydney to Cape Town
1 GB	\$1.60 / mo	\$4.41 / mo	\$2.94 / mo
10 GB	\$16.02 / mo	\$44.10 / mo	\$29.40 / mo
100 GB	\$160.20 / mo	\$441.00 / mo	\$294.00 / mo

excludes committed-use discounts, NAT gateway, Transit Gateway,
PrivateLink and VPC endpoint charges. Every figure depends on
open item L4-01-004.

WHAT EACH SHAPE FORCES ACROSS A BOUNDARY

A

T3 and T4

the arbiter holds no stream assets, so it carries
meta-group control traffic only.

Small and close to constant (L4-01-030).

V

T3, T4 and T5

mirrors and gateway message flow are the variable part.

How much is mirrored is the lever that matters.

H

T5

only au-to-hub bytes cross a region

The hub sits in af-south-1; its placement sets the bill.

Evidence note (L4-01-031)

Demo 03 ran seven topologies as bare nats-server processes on one Mac. It measured quorum and NATS behaviour. It did NOT measure AWS latency, AWS networking, real availability or cost.

Every AWS number on these sheets is a published list price or a published region fact, not a measurement.

Pick the failure first, then read the bill. The failure boundary decides the shape; the shape decides how many bytes cross a border; the rate card turns that into money. **Built today:** nothing. **Measured:** nothing on this sheet — every price is a published list price. **Not shown:** real volumes, which are open item L4-01-004.

SCOPE

One question. Which failure is bought, and what does a byte cost?
Placement is sheet 1, shapes are sheet 2.

DIRECTION

Directional as at 2026-09-18. An initial estimate, not a cost forecast.

SEPARATE

A shape that answers one failure boundary does not answer the
others (L4-01-018).

REGION vs COUNTRY

Sydney and Melbourne are two AWS Regions in one country. Two
regions is not two jurisdictions (L4-01-020).

BIAS

An arbiter in Melbourne biases the vote toward Australia. In T3,
Australia then holds 4 of 7 (L4-01-021).

BOTH ENDS

Cross-zone traffic is charged at both ends. That is why a GB across
a zone costs \$0.02, not \$0.01 (L4-01-026).

ONE WAY

Cross-region traffic is charged on the sending side only. Inbound is
free, so direction changes the bill (L4-01-027).

LEVER

At these volumes the three-zone spread is cheap. The lever is how
much is mirrored, not how many zones are used.

OPEN

Actual GB per day per cell is not known. Every figure here is a
shape, not a forecast (L4-01-004, owned by business and
architecture; L4-01-039).

LEGEND

 Green boundary — physical geography: zone, cell, region, country

 Amber panel — a rule or a rate that governs the sheet

 Amber boundary — cost, rates and rules that govern the sheet

 Red panel — a warning or a limit on the evidence

 Dashed grey tile — proposed for V3, not built

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WORDS ON THIS SHEET

AWS Amazon Web Services

AZ availability zone, one or more data centres with
their own power

cell one region's complete runtime

region a named AWS place holding several zones

opt-in a region an account must enable before
use

NAT network address translation

quorum / **majority** more than half the voters

arbiter a voter holding no stream data

mirror a copy of one stream in another place

replica one more copy of the same thing

GB gigabyte

NATS the messaging system; a product name, not
an abbreviation

NAT network address translation

VPC virtual private cloud, a private AWS network